VISHWANI SINGH

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Summary

Applied AI scientist for physiological signals in clinical populations. At Mass General I built the evaluation layer for IRB-governed patient movement and used it to find where the field's pretrained models break down on patients. Four years across production machine learning engineering and healthcare research.

Education

Northeastern University, Boston, MA, USA – MS, Artificial Intelligence · GPA 4.0/4.0 Jan 2025 – Dec 2026 (expected)

Thesis: *Do EEG Representations Transfer? A Systematic Benchmark Across Sessions, Subjects, Tasks, Paradigms, and Modalities*

Institute of Engineering and Technology, Lucknow, India – B.Tech, Information Technology · GPA 8.52/10 2017 – 2021

Experience

Massachusetts General Hospital – Applied AI Scientist, Co-op (OASIIS Lab) · Boston, MA **Jan 2026 – Sep 2026**

- Designed and led the first benchmark of movement foundation models (MOMENT, UniMTS, ImageBind, SSL-Wearables; six backbone configurations) on an IRB-governed clinical motion cohort (40 patients, 20 controls), across three tasks: injury detection, task recognition within healthy subjects, and healthy-trained task recognition tested on patients, spanning 4 fine-tuning regimes (frozen linear probe to full fine-tune), 3 sensor placements, randomly initialized counterparts for every backbone, and demographic-only baselines.
- Found healthy-population pretraining to be task-dependent. It improved injury detection for all 6 backbones (0.91 subject-level ROC-AUC, vs 0.80 for an age-only baseline on the same participants) but degraded 6-class task recognition for 4 of 6, where handcrafted features beat every pretrained model.
- Quantified the healthy-to-patient generalization gap: every backbone degraded when healthy-trained task recognition was tested on patients, and where pretraining helped at all it helped healthy subjects more than patients, evidence that a healthy-only prior carries healthy-only blind spots.
- Showed fine-tuning regimes and placements outweighed model family: a frozen linear probe matched or exceeded full fine-tuning for 5 of 6 backbones, and head-vs-left-hand placement shifted ROC-AUC by 0.165, wider than the spread across all four model families.
- Built the evaluation harness end to end: multi-GPU CUDA training on MGH's Slurm enclave, 5-seed replication, and leakage-safe subject-level splits, with model- and sensor-agnostic interfaces so a collaborating site's dataset enters as a single adapter evaluated on the same three benchmarks.

Massachusetts General Hospital and Northeastern University – Graduate Research Assistant **Mar 2025 – Dec 2025**

Supervisors: Prof. Aarti Sathyanarayana (NEU) & Dr. Christopher Bono (MGH)

- Designed the modeling and evaluation layers of a modular three-layer framework for XR-based kinematic assessment (XR task delivery and capture / model backend / reporting with reference baselines and channel attribution) of shoulder pathology: 72 deep-learning configurations across 3 architectures (CNN, RNN, Transformer), 6 DASH-derived tasks, and 4 clinical paradigms on 18 channels of 6-DoF motion from 40 patients and 20 controls, under nested leakage-safe LOSO-CV.
- Established the demographic, task-duration, and classical-ML baselines: logistic regression on age alone matched every deep model (0.750–0.810 balanced accuracy), while integrated gradients and channel ablation identified headset vertical position as the dominant channel and control misclassification tracked age (rho=0.46, p=0.042), establishing that the kinematic signal cannot be separated from participant age in this cohort.
- Designed a class-conditional hidden Markov model pipeline for generative phenotyping of XR-derived movement, fitting per-group Gaussian HMMs over 18 kinematic channels at 50 Hz and classifying held-out subjects by likelihood ratio under LOSO-CV across four diagnostic paradigms.
- Validated the latent states as movement phases rather than artifacts (state boundaries aligned with annotated task events 2.7–3.7x above chance) and used them to surface task-dependent compensation that a class label hides: patients entered more distinct states than controls during one DASH-derived task (3.8 vs 3.2, p=0.005) but froze during another, where 60% held a single state for half the trial versus 5% of controls.

Northeastern University – Teaching Assistant · Boston, MA **Sep 2025 – Dec 2025**

Supervisor: Prof. Iraklis Tsekourakis

- Teaching Assistant for CS 5800: Algorithms, assisting with in-person office hours, student support, course teaching, assignment grading, planning and examination invigilation.

Indian Institute of Technology, Delhi – Senior Research Fellow · Delhi, IN **Sep 2023 – Aug 2024**

Supervisor: Prof. Tapan K. Gandhi

- Designed and led a graph-theoretic resting-state EEG functional-connectivity pipeline (32-channel, 500 Hz; average re-referencing, 0.01–45 Hz band-pass, 4 s epoching at 50% overlap, Welch power spectral density, Morlet wavelet decomposition, multitaper phase-lag-index connectivity) to quantify effects of 20 HD-tDCS sessions across theta/alpha/beta bands in an IRB-approved MDD cohort at AIIMS Delhi.

- Contributed a modified adaptive binarization algorithm (median + k-sigma thresholding validated against 100 edge-matched Erdos-Renyi nulls and 1000 permutations), identifying increased beta-band small-worldness and DLPFC hubness as candidate network biomarkers of treatment response.

GeeksforGeeks – Software Engineer II (MLE) · Noida, IN **Sep 2021 – Aug 2023**

- Built and optimized scalable backend infrastructure for ML-driven search and ranking (AWS OpenSearch + Redis) serving 1M assets; trained BERT-based query-routing classifiers across 10+ academic domains, cutting human escalations 35% at 50K+ queries/month.
- Built an embedding-based code-retrieval system and a Neo4j graph recommender matching code errors via AST analysis adopted by 100K+ users with a 25% higher first-attempt solution rate; shipped to production as Django/DRF REST APIs.

Technical Skills

Foundation Models & Representation Learning: PyTorch, self-supervised pretraining, linear probing & full fine-tuning, few-shot transfer, Transformers, Mixture-of-Experts, TCN, CNN, LSTM / GRU, HMM, Hugging Face, W&B

Time Series, Signals & Multi-Sensor Data: Multivariate time-series modeling, IMU & EEG signal processing, epoching / windowing, spectral analysis (PSD, frequency-band decomposition), normalization & alignment, graph-theoretic connectivity (phase lag index, small-worldness, hubness), MNE-Python, braindecode, BIDS-EEG, scikit-learn, Pandas, NumPy

Evaluation & Interpretability: Leakage-aware subject-level splits, LOSO cross-validation, multi-seed replication, ablation studies, permutation testing, feature importance, embedding-space & cluster analysis, explainable AI for clinical decision support

Engineering & Infrastructure: Python, SQL, Linux, CUDA, multi-GPU training, HPC / Slurm, Docker, Git, GitHub Actions, AWS (S3, EC2, OpenSearch), Redis, Django / DRF, REST APIs, RAG

Clinical & Research: Digital biomarker discovery, clinical & translational AI, IRB-governed clinical data workflows, XR-based movement assessment, biomechanical analysis

Publications and Presentations

- **V. Singh** et al. "Evaluating and Adapting Movement Foundation Models for Pathology Sensitivity and Task Invariance." 2026. [In Preparation for IMWUT]
- **V. Singh** et al. "Biomechanical Phenotyping and Classification of Shoulder Pathology with Hidden Markov Modeling and Extended Reality." *IEEE JBHI*, 2026. [Under Review]
- N. Miner, **V. Singh**, et al. "Wearable Inertial Sensors and Machine Learning for Clinical Assessment of Human Movement: A Systematic Review of Health-Based Applications." *Journal of NeuroEngineering and Rehabilitation (JNER)*, 2026. [Under Review]
- **V. Singh** et al. "Standardized Extended Reality Task Delivery and Kinematic Classification of Shoulder Pathology: Proof-of-Concept Study." *JMIR XR & Spatial Computing*, 2026. [Minor Revision]
- **V. Singh**, R. Verma, S. Shriyam, and T. K. Gandhi, "Evaluating tDCS Intervention Effectiveness via Functional Connectivity Network on Resting-State EEG Data in Major Depressive Disorder." 2024. [Research presentation]

Projects

Subject-Invariant EEG Decoding *(NeurIPS 2025 EEG Challenge)* - Built a subject-invariant decoder predicting reaction time from 128-channel EEG with no per-subject calibration, combining spatial filters, multi-temporal convolutions, and a Mixture-of-Experts head with custom load-balancing losses; RMSE 0.3609 vs. the 0.3914 EEGSimpleConv baseline. Extended into thesis work on cross-subject and modality transfer.

Transformer-Based Soccer Action Spotting *(CS 7150 Deep Learning)* - Temporal action-spotting pipeline (positional encoding + 8-head attention pooling) on SoccerNet v2 (471 matches, 716+ hours) using ResNet embeddings at 2 fps. 34.7% validation mAP, 2x the average-pooling baseline (17.8%).

Human-Centered AI Dashboard for Clinical XR *(CS 5170 AI for Human-Computer Interaction)* - Led user research with clinicians at Mass General Hospital (12 surveys, 4 semi-structured interviews, 5.5 hrs contextual observation), then built a Gemini-grounded RAG assistant over clinical guidelines and motion-assessment protocols and a three-stage XAI framework (Glance / Scan / Investigate) for clinician decision support.